

What the competition was, where we landed, and the road to top 10

A plain-English retrospective on the Zindi **Multilingual Health Question Answering in Low-Resource African Languages** challenge — and an honest comparison against the published 11th-place solution.

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One-line thesis	We engineered a great car and never drove it. The winners drove a simpler one — and the fuel was retrieval .

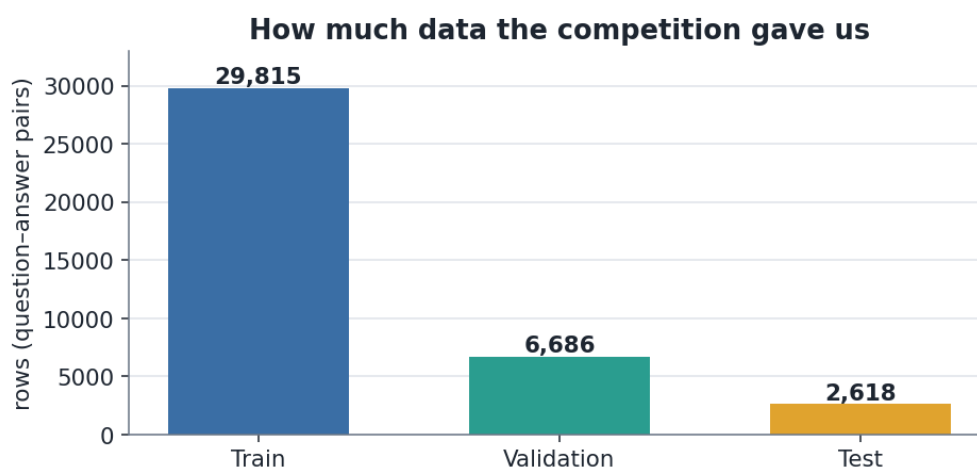
How to read this report

Part A explains the competition with no jargon. Part B is an honest look at what we built. Part C compares us to the 11th-place finisher and names what we missed. Part D is the concrete plan to reach top 10. Numbers shown as ROUGE are **measured**; any LLM-judge figure is clearly labelled **estimated**.

Part A · The competition, in plain English

Imagine a free health helpline for young people across Africa — covering questions about the body, relationships, pregnancy, and sexually transmitted infections. People write in their own language. The machine has to write back a good answer **in that same language**. That is the whole competition: read a health question, write a helpful answer in the right language.

Eight 'subsets' were in play — the same kind of health questions asked in different languages and countries: English (in Uganda, Ghana, Kenya, Ethiopia), plus **Luganda** (Uganda), **Kiswahili** (Kenya), **Akan/Twi** (Ghana) and **Amharic** (Ethiopia).



Test answers are hidden — we submit predictions for 2,618 questions and Zindi grades them.

Figure 1. We were handed ~30,000 example Q&A; pairs to learn from, and 2,618 test questions whose answers are hidden.

A surprising fact shaped everything: **despite the 'African languages' billing, about 61% of the data is English**. A quarter of it is English from Uganda alone. So the single biggest way to score points was simply to be very good at the English questions first.

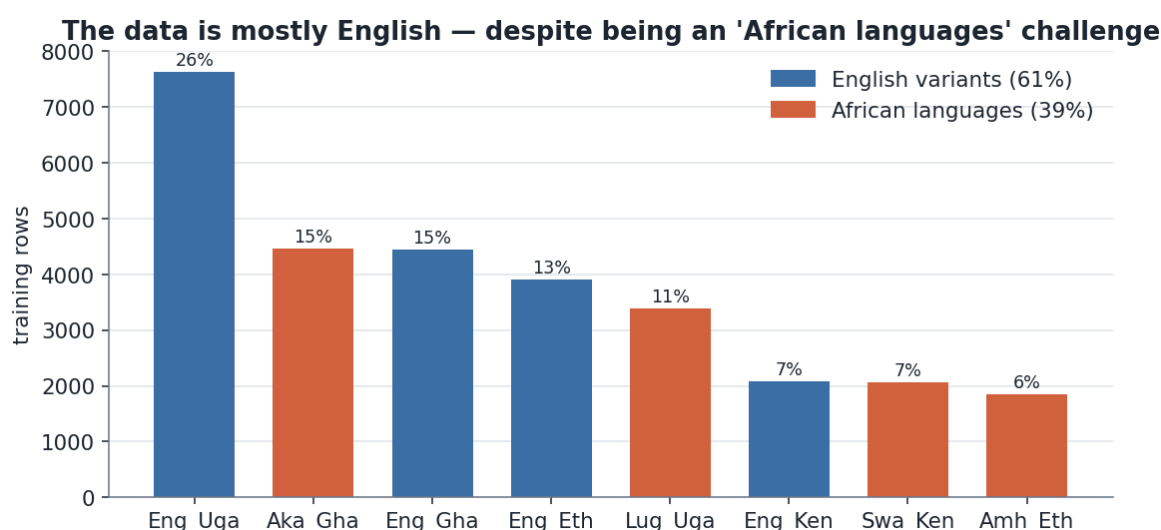
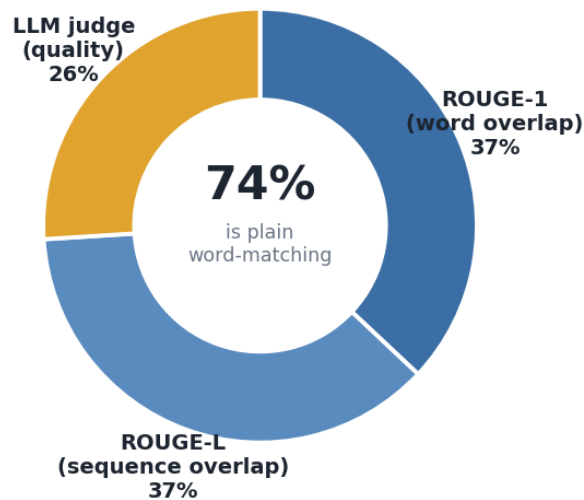


Figure 2. The language mix. Blue = English variants (~61%), orange = African languages (~39%).

How answers are graded. Your answer is compared to a reference 'model answer'. Most of the score — 74% — is just *word overlap* with that reference (a metric called ROUGE). The remaining 26% is an AI 'judge' rating quality. A third metric, AfroLM-BertScore, sounds important but is worth **zero** on the leaderboard.

How answers are scored



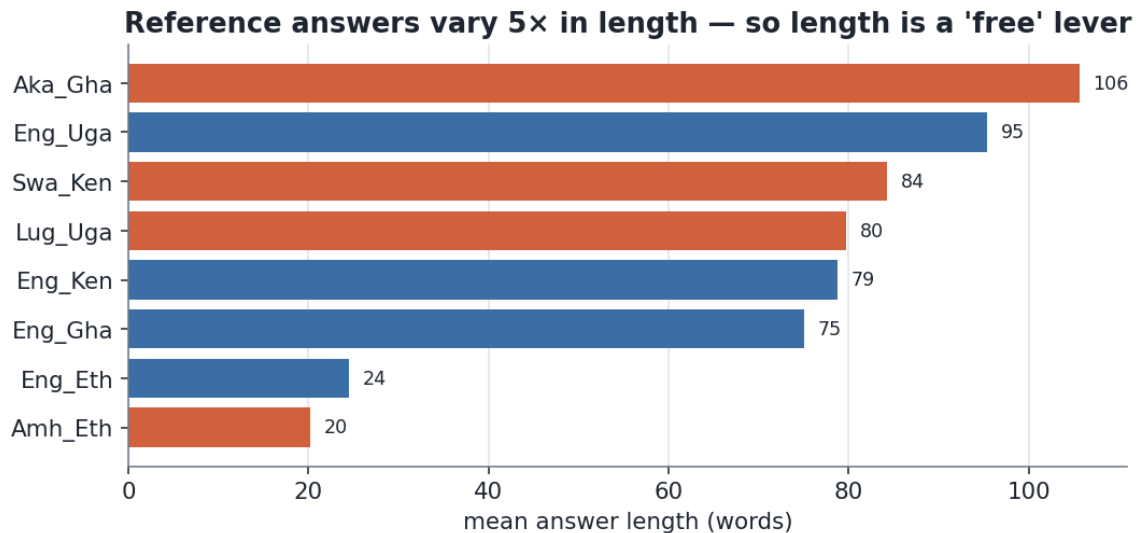
AfroLM-BertScore = 0% on the leaderboard. Matching the reference wording matters most.

Figure 3. The scoring recipe. Because 74% is word-matching, the goal is to produce the reference's words — not the prettiest essay.

Why this matters

If the score is mostly word-overlap, then an answer that already exists in the training data — and matches a new question — is worth a lot. Hold that thought; it is the whole story of Part C.

One more lever: **length**. ROUGE is an F1 score, so an answer that is too short or too long both lose points. Reference answers range from ~20 words (Amharic) to ~106 words (Akan) — a 5× spread. Matching each language's typical length is nearly-free points.



ROUGE is an F1 score: too-short or too-long answers both lose points. Match each language's typical length.

Figure 4. Typical answer length by subset. Tuning output length per language is one of the cheapest ways to gain ROUGE.

Part B · What we built — and where we actually landed

Here is the uncomfortable truth, stated plainly. We built an **excellent engineering scaffold**: clean train/validation/held-out splits, a faithful copy of the scoring metric, three model configurations with QLoRA fine-tuning recipes, an Apple-Silicon inference path, a reproducibility harness, and pages of strategy docs.

...but the engine was never started

Our experiment log contains exactly one row: the bootstrap commit. All ten of our planned experiments are still marked 'Pending'. **We never fine-tuned a model, never logged a single scored experiment, and never made one submission to Zindi.** Our only real measurement was a crude retrieval probe.

Our gap in one picture: we built a lot, and ran almost none of it

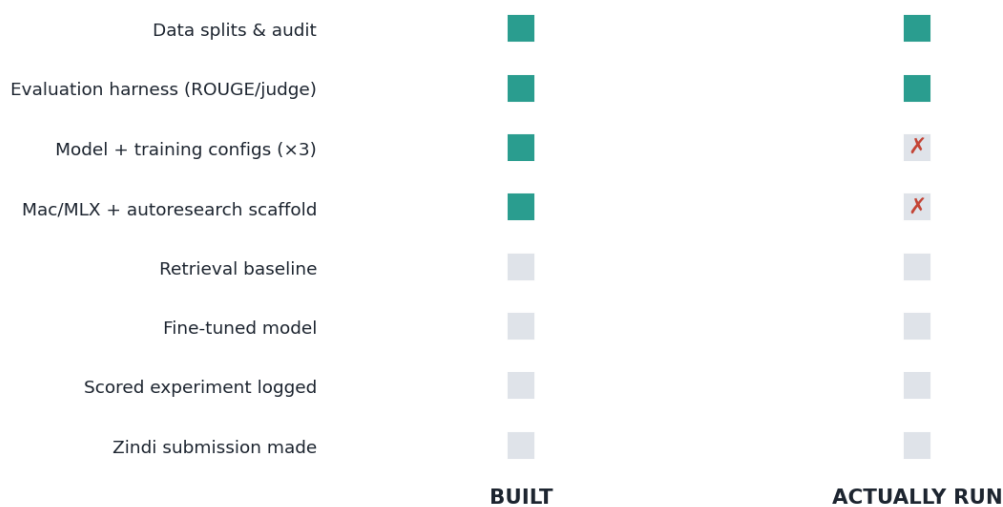


Figure 5. Built versus actually run. Green squares on the left, almost empty on the right. The whole story of our result is in that second column.

This is not a tooling failure — the tools are good. It is an **execution** failure: we polished the workshop instead of shipping a car. And the cost of that is exact and measurable, which Part C makes painfully clear.

Part C · The 11th-place solution — and the gap

The 11th-place finisher (Koleshr) wrote up their approach. They were refreshingly honest: they only got serious in the **final four days**. They did not test many models or fancy decoders. They did one thing well, and it was enough for 11th place.

Their core insight: the training data is also a knowledge base. Many test questions are near-duplicates or paraphrases of questions already answered in the training set. So instead of always generating an answer, you can *retrieve* the closest known question and reuse its answer. Because the score is mostly word-overlap, a real reference answer scores very high.

Retrieval alone — no fine-tuning — was already strong (their numbers)

Using BGE-M3 (a strong multilingual embedding model) to retrieve the nearest answer: **ROUGE-1 = 0.5548, ROUGE-L = 0.4823, LLM-judge = 0.7379** → a combined score of about **0.576**. That is a serious score from *copying* the right existing answer. Their best fine-tuned and RAG+fine-tuned models then reached ~0.66.

We reproduced this with our own data. Retrieving the nearest training answer per language on our sealed held-out slice with BGE-M3 scores **ROUGE-1 = 0.560, ROUGE-L = 0.491** (ROUGE-only combined ≈ 0.389, before the judge term). Compare that to the crude token-overlap probe we had been quoting — ROUGE-1 0.404, ROUGE-L 0.350. **Swapping in proper dense embeddings lifts ROUGE-1 by roughly ++0.16 for essentially no extra work.**

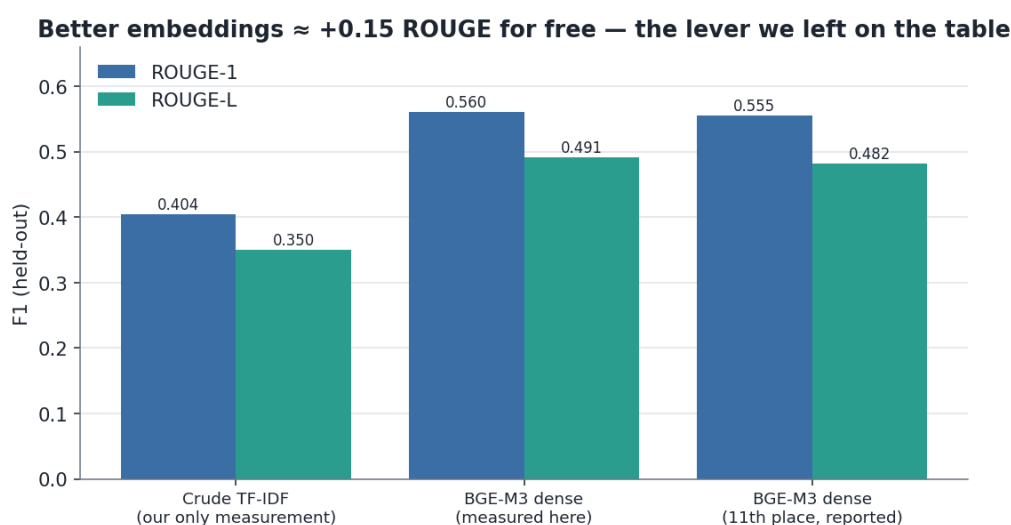


Figure 6. The same idea, three embedders. Better embeddings are worth ~+0.15 ROUGE for free — the lever we identified and never pulled.

And here is the part that stings. **Our own intelligence document already said all of this.** Months ago, `COMPETITION_INTEL.md` measured retrieval as strong, flagged that 41.6% of the English-Ethiopia questions were near-identical to training questions, and explicitly recommended a *retrieval + generation hybrid* — the exact shape of the winning approach. We wrote the winning idea down and then filed it away.

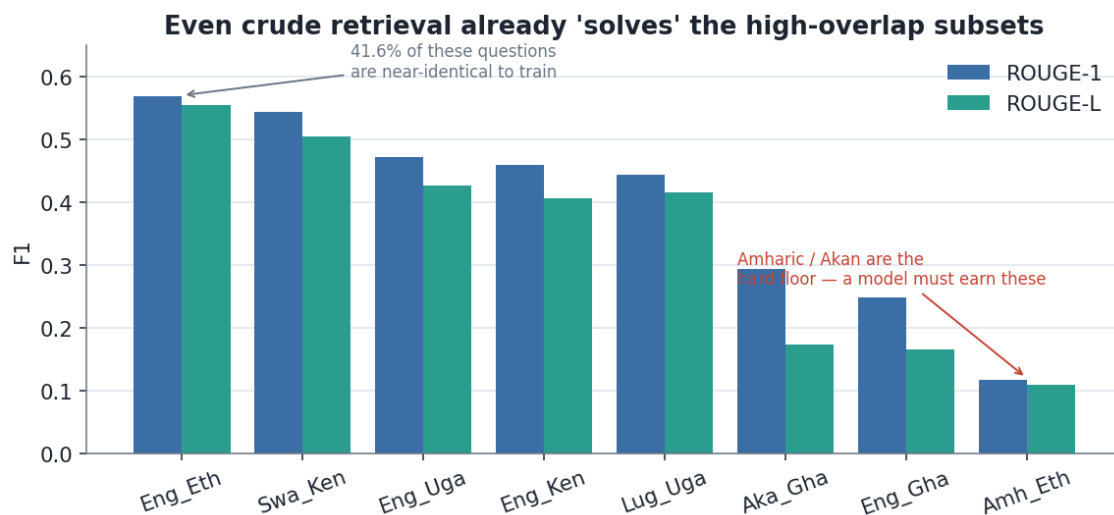


Figure 7. Even our crude retrieval already 'solved' the high-overlap English subsets; Amharic and Akan are the genuine hard floor.

Stacking the approaches on one ladder makes the gap concrete:

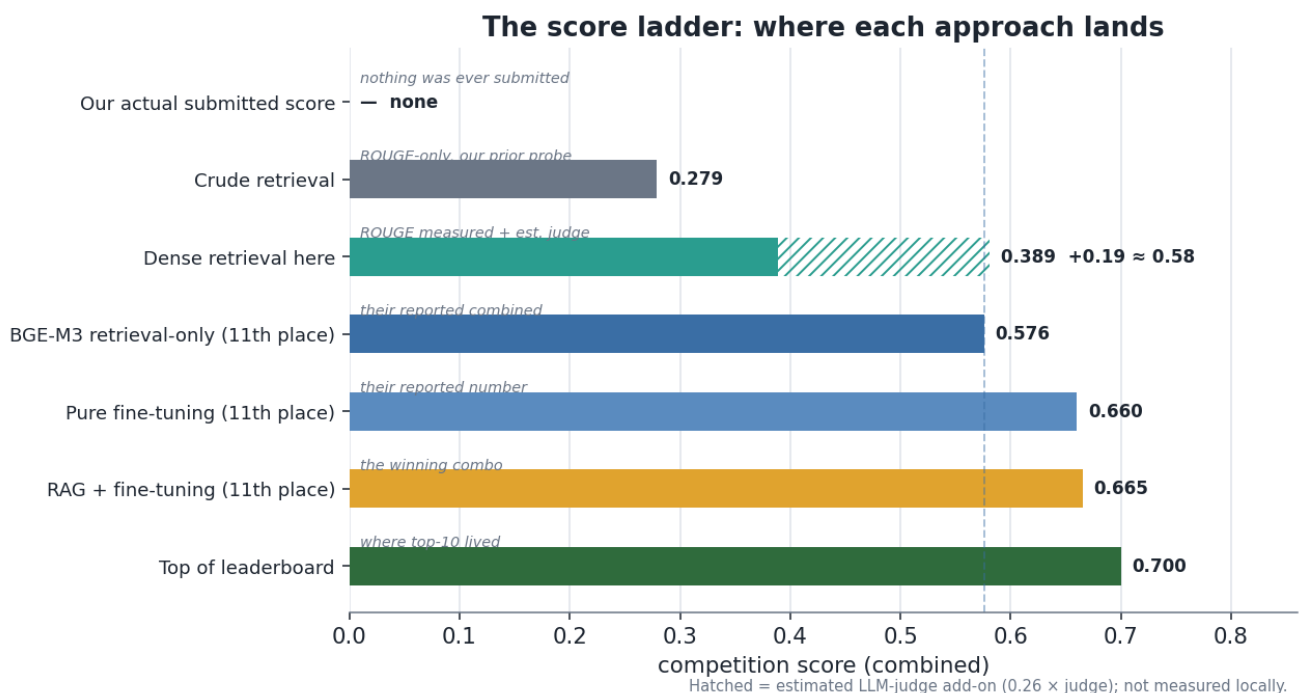


Figure 8. The score ladder. Our actual submitted score is the bottom rung — there isn't one. Retrieval alone would have cleared most of the climb.

What we missed

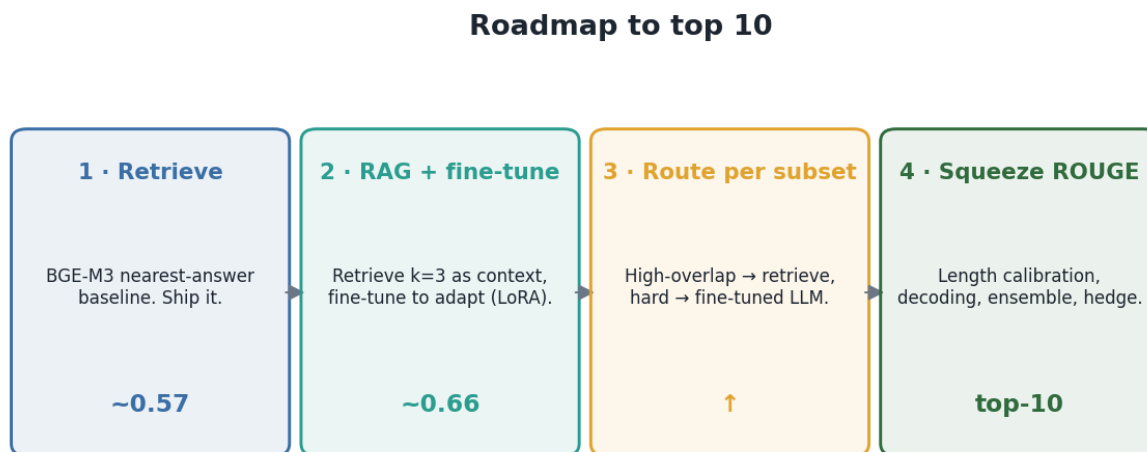
- **Retrieval as a knowledge base.** The single highest-leverage idea — and it needed no GPU training to start scoring.
- **Dense embeddings.** We measured a weak crude-TF-IDF lower bound (0.40) and believed it; a proper embedder (BGE-M3) is far stronger.
- **RAG + fine-tuning.** Retrieve similar examples, put them in the prompt, and fine-tune the model to adapt them — the combination that reached ~0.66.
- **Shipping anything at all.** A finisher who started 4 days out still placed 11th. Submitting beats perfecting.

Where we went wrong

- We invested in **infrastructure** (three model configs, a Mac path, a reproducibility harness) before running one real experiment.
- We **dismissed retrieval**, half-conflating 'don't chase external data' with 'don't reuse the labelled pool' — and threw out the most valuable signal in the data.
- We **trusted a crude measurement** instead of spending an afternoon on a proper one.
- Metric confusion (optimising the wrong weights, including a zero-weight metric) burned time early — documented in our own LESSONS.md.
- We never closed the loop: **no submission, no leaderboard feedback, no iteration.**

Part D · The roadmap to top 10

The competition has closed, so this is a **practice plan** — the repeatable playbook that would have put us in contention, and will next time. The rule above all others: **each phase ships a scored submission before the next begins**. Measure, never estimate.



Each phase ships a scored submission before the next begins — measure, don't estimate.

Figure 9. Four phases, each ending in a real submission.

Phase 1 — Retrieve, and ship it (target ≈ 0.57)

Build the BGE-M3 nearest-answer baseline (the very script in this repo). Retrieve from train + validation, per language, and submit. This locks a strong floor on day one and proves the whole pipeline end-to-end — exactly what we never did.

Phase 2 — RAG-enriched fine-tuning (target ≈ 0.66)

For each training row, retrieve the k=3 most similar examples (excluding itself), put them in the prompt as context, and fine-tune a strong multilingual model (Aya-Expansive / AfriqueLlama / Sunbird-Sunflower) with LoRA so it learns to *adapt* the retrieved text rather than copy it. Serve with vLLM. This is the 11th-place winner's exact recipe.

Phase 3 — Route per subset (climb the board)

Some subsets (high-overlap English) are won by retrieving directly; others (Amharic, Akan) need the fine-tuned model to generate. Build a simple router: if the nearest retrieved question is similar enough, return its answer; otherwise generate. Tune the similarity threshold on the held-out slice **by ROUGE** — never by an unscored metric.

Phase 4 — Squeeze the last ROUGE (into top 10)

Per-subset length calibration, a short decoding sweep, a light two-model ensemble, and a final hedged pair (best-public vs best-held-out). These are small, compounding gains on top of a strong base.

The one-sentence lesson

We did not lose on ideas — our own notes held the winning strategy. We lost on **execution and humility**: trust the data, build the simplest thing that scores, submit it, and iterate from a real number.

Appendix · Exact numbers & references

Scoring formula (from `src/afro_health_qa/evaluation/combined.py`)

$\text{combined} = 0.37 \cdot \text{ROUGE-1 F1} + 0.37 \cdot \text{ROUGE-L F1} + 0.26 \cdot \text{LLM-judge}$ (AfroLM-BertScore = 0.00).

Retrieval, measured here vs reported

	ROUGE-1	ROUGE-L	ROUGE-only comb.	+ judge
Crude TF-IDF (our prior probe)	0.404	0.350	0.279	—
Dense (bge-m3), measured	0.560	0.491	0.389	not run
BGE-M3, 11th place (reported)	0.5548	0.4823	0.384	0.576

ROUGE-only comb. = $0.37 \cdot R1 + 0.37 \cdot RL$. The '+ judge' column adds the 0.26 LLM-judge term; we did **not** run the judge locally, so our dense row leaves it blank and the 11th-place row uses their reported judge value. Note our local ROUGE uses `rouge_score` with `stemmer=True`, which can differ slightly from the leaderboard's whitespace tokenizer — treat absolute values as indicative, the gaps as real.

Data at a glance

Train 29,815 · Validation 6,686 · Test 2,618 · sealed held-out 1,491. Eight subsets; ~61% English (Eng_Uga alone ~26%). Answer length 20–106 words by subset.

Key files

Charts: `scripts/build_report_charts.py` · Retrieval baseline: `scripts/retrieval_baseline_bge.py` · This report: `scripts/build_report_pdf.py` · Source-of-truth intel: `autoresearch_nlp/COMPETITION_INTEL.md` · Scoring: `src/afro_health_qa/evaluation/combined.py`.